MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, PLUG, RIGHT ANGLE, DECADE INCREMENT 20 THROUGH 110 AND 128 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.050 SPACING)

This specification is approved for use by Rome Air Development Center, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

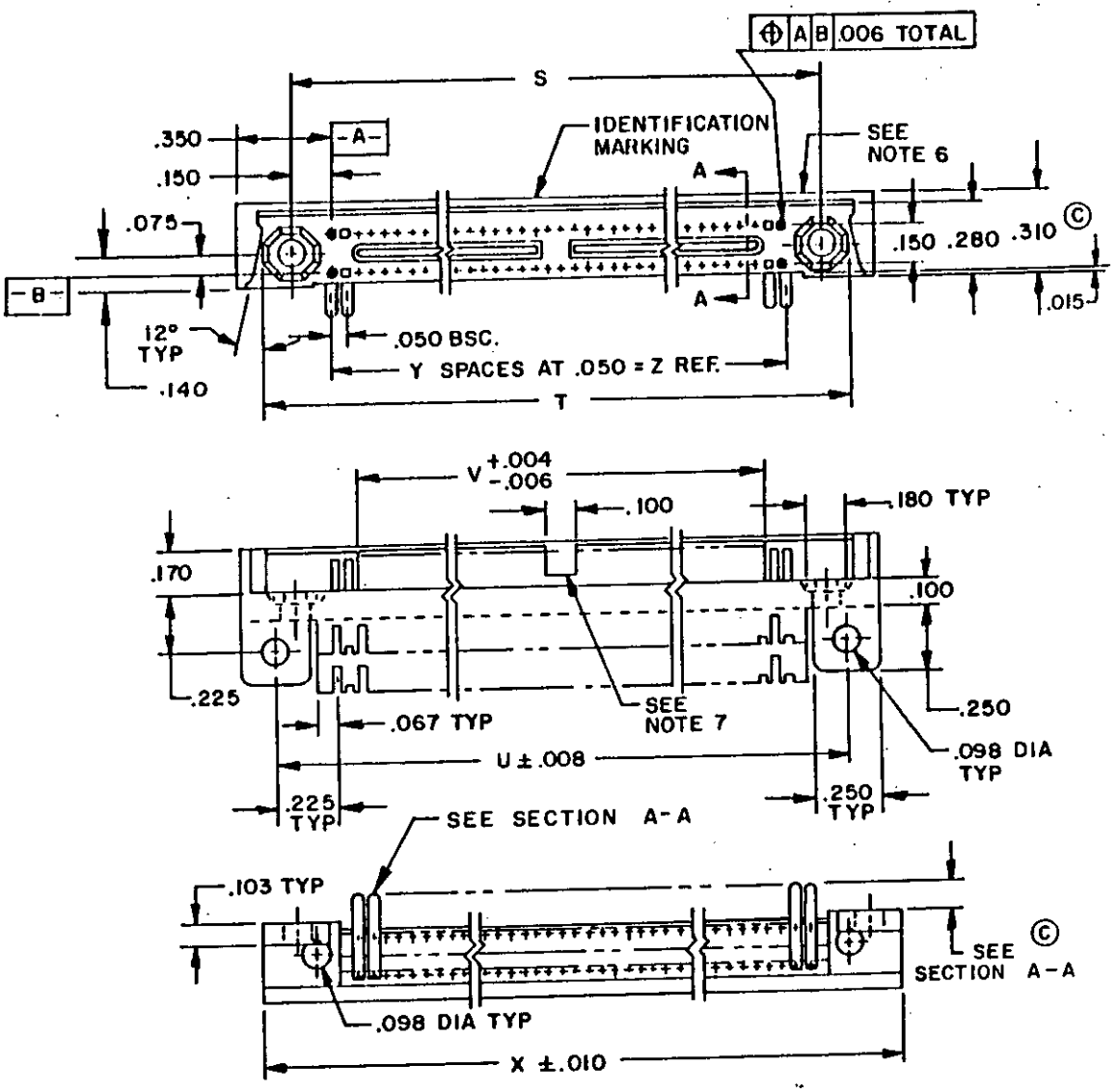
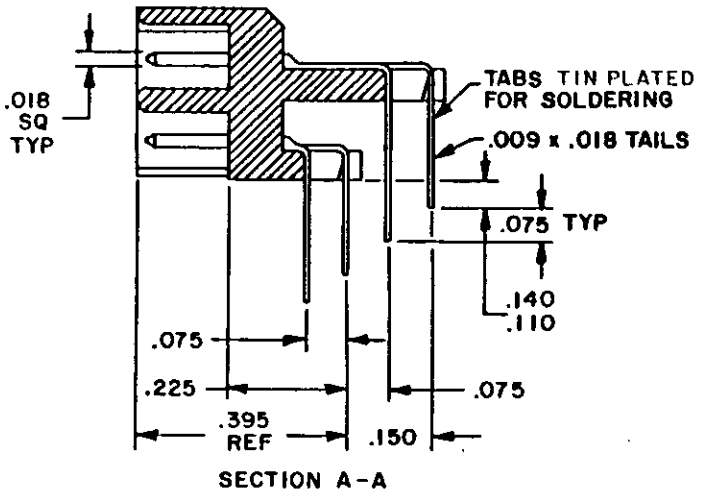


FIGURE 1. Connectors, plug .050 (1.27 mm) spacing.

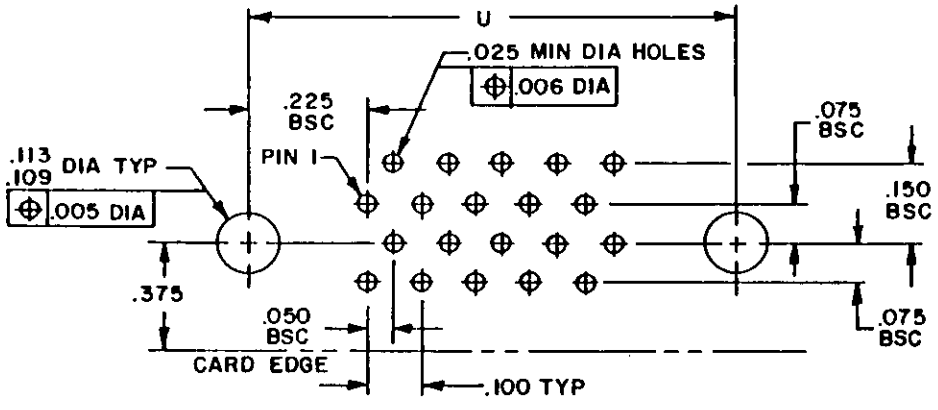
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SECTION A-A



RECOMMENDED PRINTED WIRING BOARD LAYOUT

Inches	mm	Inches	mm	Inches	mm
.004	.10	.075	1.90	.170	4.32
.005	.13	.098	2.49	.180	4.57
.006	.15	.100	2.54	.225	5.72
.008	.20	.103	2.62	.250	6.35
.009	.23	.109	2.77	.280	7.11
.010	.25	.110	2.79	.310	7.87
.018	.46	.113	2.87	.350	8.89
.050	1.27	.140	3.56	.375	9.52
.067	1.70	.150	3.81	.395	10.03

FIGURE 1. Connectors, plug .050 (1.27 mm) spacing - Continued.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals.
4. These connectors mate with connectors specified in MIL-C-55302/119(USAF).
5. Pads suitable for printed wiring board support are optional.
6. Numbers indicating end positions stamped this side only.
7. Void is not present on 20 through 50 position.

FIGURE 1. Connector, plug .050 (1.27 mm) spacing - Continued.

TABLE I. Dash numbers and dimensions.

Dash number	No. of contacts	Dimensions							Contact identification numbers
		S	T	U	V	X	Y	Z	
01	20	.750 (19.05)	.970 (24.64)	.900 (22.86)	.290 (7.37)	1.150 (29.21)	9	.450 (11.43)	1-10
02	30	1.000 (25.40)	1.220 (30.99)	1.150 (29.21)	.540 (13.72)	1.400 (35.56)	14	.700 (17.78)	1-15
03	40	1.250 (31.75)	1.470 (37.34)	1.400 (35.56)	.790 (20.07)	1.650 (41.91)	19	.950 (24.13)	1-20
04	50	1.500 (38.10)	1.720 (43.69)	1.650 (41.91)	1.040 (26.42)	1.900 (48.26)	24	1.200 (30.48)	1-25
05	60	1.750 (44.45)	1.970 (50.04)	1.900 (48.26)	1.290 (32.77)	2.150 (54.61)	29	1.450 (36.83)	1-30
06	70	2.000 (50.80)	2.220 (56.39)	2.150 (54.61)	1.540 (39.12)	2.400 (60.96)	34	1.700 (43.18)	1-35
07	80	2.250 (57.15)	2.470 (62.74)	2.400 (60.96)	1.790 (45.47)	2.650 (67.31)	39	1.950 (49.53)	1-40
08	90	2.500 (63.50)	2.720 (69.09)	2.650 (67.31)	2.040 (51.82)	2.900 (73.66)	44	2.200 (55.88)	1-45
09	100	2.750 (69.85)	2.970 (75.44)	2.900 (73.66)	2.290 (58.17)	3.150 (80.01)	49	2.450 (62.23)	1-50
10	110	3.000 (76.20)	3.220 (81.79)	3.150 (80.01)	2.540 (64.52)	3.400 (86.36)	54	2.700 (68.58)	1-55
11	128	3.450 (87.63)	3.670 (93.22)	3.600 (91.44)	2.990 (75.95)	3.850 (97.79)	63	3.150 (80.01)	1-64

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REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figure 1.

Material:

© Body: In accordance with MIL-C-55302.

© Plating: Contact plating shall be gold in accordance with MIL-C-55302 over nickel in accordance with MIL-C-55302 on the contact engagement area for .100 inch minimum length. The termination area plating shall be tin lead in accordance with MIL-C-55302 over nickel in accordance with MIL-C-55302 for .300 inch minimum length. The remaining areas of the contact shall be nickel plated in accordance with MIL-C-55302.

Contact identification: See figure 1.

Contact rating: 1.5 amperes per contact, maximum.

Operating temperature: -65°C to +125°C.

Keying: See MIL-C-55302/31.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .30 and the withdrawal force in pounds shall be a minimum of .03 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: The average contact resistance of all contacts measured shall not exceed .025 ohm, and no individual contact pair shall have a resistance exceeding .030 ohm.

Contact retention: .75 pound, minimum.

Dielectric withstanding voltage:

Sea level: 600 volts rms.

High altitude: 150 volts rms.

Solderability: Each connector shall be subjected to method 208 of MIL-STD-202 (solder termination only) (applicable to qualification and retention of qualification only).

Part number: M55302/117 - (dash number from table I).

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

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Air Force - 17

Review activities:
Air Force - 11, 85, 99
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